Year 1 Science Student Notes

Exploring Science 7 | Chapters 7Aa to 7Ad

Learn the key ideas about living things, organs, tissues, microscopes and cells. Use the page references to find each topic in your book. Numbered book answers follow the notes.

7Aa Doctors past and present

Book page 5

Symptoms are changes or problems that suggest an illness. Examples include a sore throat, a runny nose, a headache and a high temperature.

Evidence is information that supports a conclusion. Doctors compare symptoms with known illnesses and may use examinations, scans, blood tests or urine tests to help identify the cause.

Ancient and modern doctors both examined patients and checked the heartbeat. Modern equipment allows doctors to see inside the body and examine samples in much greater detail.

7Aa Life processes

Book pages 6 and 7

An organism is a living thing. Living things carry out seven life processes. Remember them using MRS GREN.

Letter	Life process	Meaning
M	Movement	Moving the whole organism or parts of it.
R	Reproduction	Producing offspring.
S	Sensitivity	Detecting and responding to changes in the surroundings.
G	Growth	Increasing in size.
R	Respiration	Releasing energy from food in cells.
E	Excretion	Removing waste made by the body's activities.
N	Nutrition	Obtaining substances needed for energy and growth.

Plants are organisms too: they grow, move parts of themselves, respond to their surroundings, reproduce and respire.

7Aa Understanding life processes

Book pages 6 and 7

Movement and sensitivity

Most animals move from place to place. Most plants stay rooted in one place but move parts of themselves, for example by growing towards light.

A stimulus is a change that an organism can detect. Light, touch, temperature and sound are examples. A response is what the organism does after detecting a stimulus.

Example: touching a hot object is a stimulus; pulling your hand away is a response.

Respiration and nutrition

Respiration releases energy from food. This energy is used for activities such as movement, growth and repair. Many organisms use oxygen during respiration.

Breathing moves air into and out of the lungs. Respiration is a chemical process in cells. Plants respire as well as animals.

Animals eat food. Plants make food by photosynthesis and absorb water and mineral salts from the soil.

Excretion

Examples of excreted wastes are carbon dioxide from respiration and urea in urine. Removing undigested food as faeces is called egestion.

Living and non living things

Movement alone does not show that something is alive. Cars and robots can move but do not carry out all seven life processes as organisms do.

Organisms	Not organisms
Cow, daffodil, goldfish, mouse, octopus, snake	Car, chair, coal, robot, rock, Sun

Coal and a wooden chair may come from once-living material, but they are not living organisms now.

7Ab Human organs

Book page 8

An organ is a body part made of different tissues working together to carry out a function. A function is the job that a structure does.

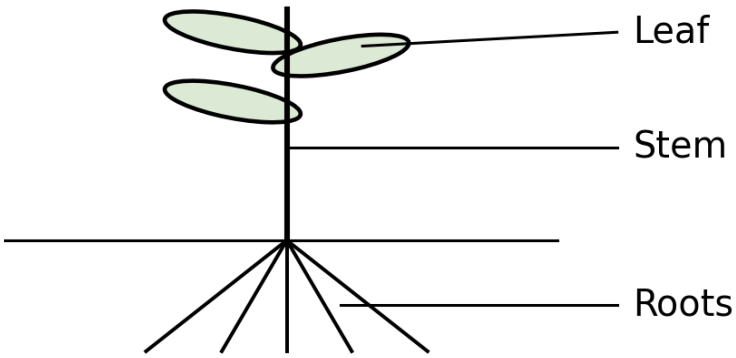
Organ or structure	Main function
Brain	Controls and coordinates the body.
Skin	Protects the body and detects changes such as touch and temperature.
Trachea or windpipe	Carries air towards and away from the lungs.
Lungs	Take oxygen into the blood and remove carbon dioxide.
Heart	Pumps blood around the body.
Diaphragm	A sheet of muscle below the lungs that helps breathing.
Oesophagus or gullet	Carries swallowed food to the stomach.
Stomach	Churns food and begins some digestion.
Liver	Makes and stores substances and breaks down harmful substances.
Small intestine	Digests food and absorbs nutrients into the body.
Large intestine	Absorbs water from the remaining undigested material.
Rectum	Stores faeces before they leave the body.
Kidneys	Remove wastes from the blood and produce urine.
Bladder	Stores urine before it leaves the body.

Blood carries oxygen and nutrients to body cells. The heart keeps this blood moving.

Do not confuse the trachea, which carries air, with the oesophagus, which carries food.

7Ab Plant organs

Book page 9



Organ	Main functions
Leaf	Absorbs light and makes food by photosynthesis.
Stem	Supports leaves and flowers and carries substances around the plant.
Root	Anchors the plant and absorbs water and mineral salts from the soil.
Storage organ	Stores food made by the plant. Examples include carrots and potatoes.

Photosynthesis

Photosynthesis is the process by which plants use light energy to make food from carbon dioxide and water. Oxygen is also produced.

Leaves contain chlorophyll, a green substance that absorbs light. Roots provide water and stems carry it to the leaves.

Carbon dioxide + water → glucose + oxygen

Light energy and chlorophyll are needed. Glucose is a sugar that the plant can use for respiration, growth or storage.

With very little light, a plant makes less food, so its storage organs may grow poorly.

7Ac Tissues

Book pages 12 and 13

A tissue is a group of cells of the same type working together. Different tissues combine to make an organ.

Cell → Tissue → Organ → Organ system → Organism

Example: heart muscle cells form muscle tissue. Muscle tissue and other tissues make the heart. The heart works with blood vessels in the circulatory system.

Tissue	Where it is found	Function
Muscle tissue	Heart and intestines	Contracts to cause movement; heart muscle pumps blood.
Fat tissue	Around organs such as the heart	Helps cushion and protect organs and stores energy.
Root hair tissue	On the outside of roots	Helps absorb water and mineral salts from the soil.
Xylem tissue	Roots, stems and leaves	Carries water and dissolved mineral salts through the plant.

Organs contain different tissues

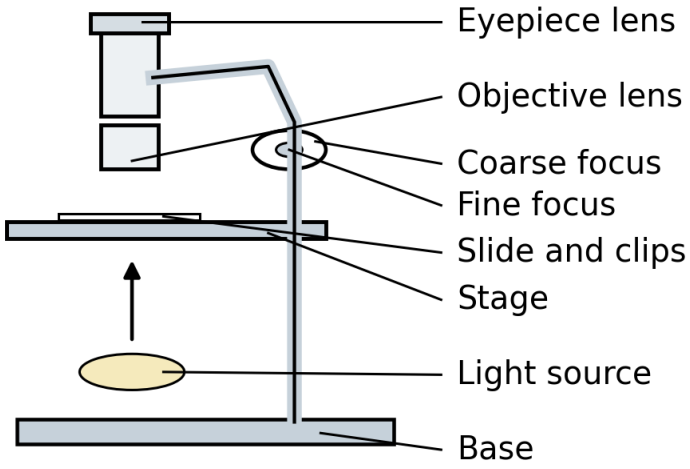
The heart contains muscle and fat tissue. A carrot is a storage root containing several tissues, including xylem and food-storage tissue.

A pathologist is a doctor who examines tissues to help identify diseases or investigate causes of death. A microscope makes small structures appear larger, so details can be seen.

A biopsy is a small sample of tissue taken from a living person for examination.

7Ac Using a light microscope

Book page 14



Part	Job
Eyepiece lens	The lens you look through; magnifies the image.
Objective lens	Magnifies the specimen; different objectives give different magnifications.
Stage and clips	Support the slide and hold it in place.
Light source or mirror	Directs light through the specimen.
Coarse focusing wheel	Makes large focus adjustments at low magnification.
Fine focusing wheel	Makes small adjustments to sharpen the image.

Start with the lowest-power objective. Centre the specimen over the stage opening and adjust the light. Focus at low power before changing to a higher-power objective.

At high power, use fine focus. Keep the lens from touching the slide. Carry the microscope with two hands and never aim its mirror at the Sun.

7Ac Preparing slides and magnification

Book page 15

A specimen is the material you want to examine. It must be thin enough for light to pass through it.

Preparing an onion tissue slide

1. Place a drop of water on a clean glass slide.
2. Use forceps to peel a very thin piece of onion epidermis.
3. Lay the tissue flat in the water. A suitable stain can make structures easier to see.
4. Lower a coverslip slowly at an angle to avoid trapping air bubbles.
5. Observe at low power, focus, and increase magnification if needed.

The coverslip holds the specimen flat, keeps it in place and slows drying. Wear eye protection when required and handle slides and coverslips carefully.

Calculating magnification

Total magnification = eyepiece magnification × objective magnification

Example: a ×10 eyepiece and a ×15 objective give ×150 total magnification.

Magnification tells us how many times larger an image appears than the real object.

Real size = image size ÷ magnification

Example: an image 40 mm wide at ×400 represents a cell 0.10 mm wide. Use the same units throughout the calculation.

If the view is dark

Check the light source, the opening beneath the specimen, the slide position and the objective position. The specimen may be too thick, or something may be blocking the light.

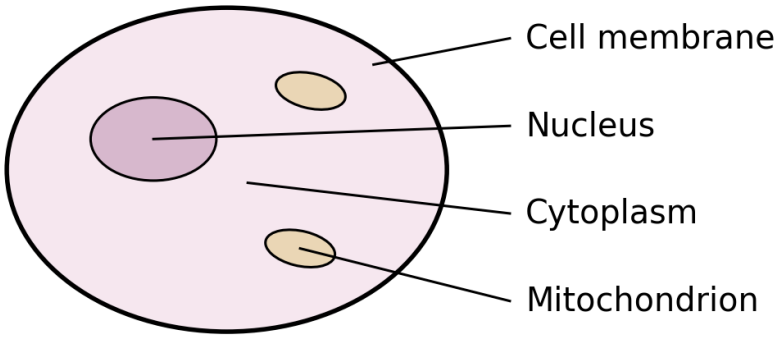
7Ad Cells

Book pages 16 and 17

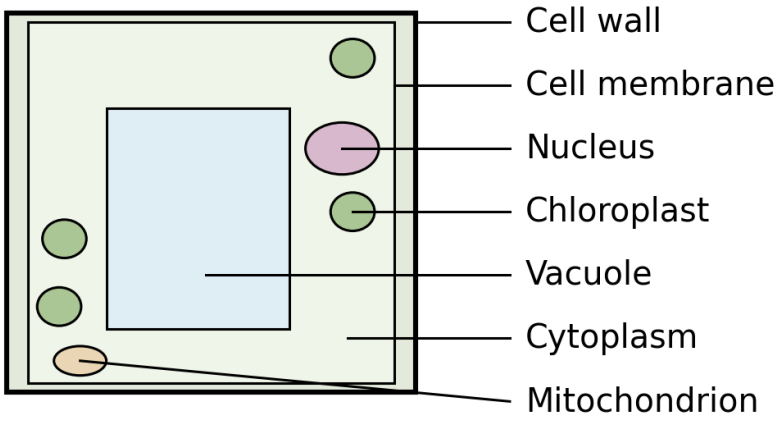
A cell is the basic unit of a living organism. Tissues are made of cells. Specialised cells have structures suited to a particular job.

Robert Hooke examined cork in 1665 and named the small box-like spaces cells. Improved microscopes later allowed scientists to see more detail.

A typical animal cell



A typical green plant cell



These are simplified diagrams. Real cells have many shapes and sizes. A typical green leaf cell has chloroplasts; many other plant cells, such as root hair cells, do not.

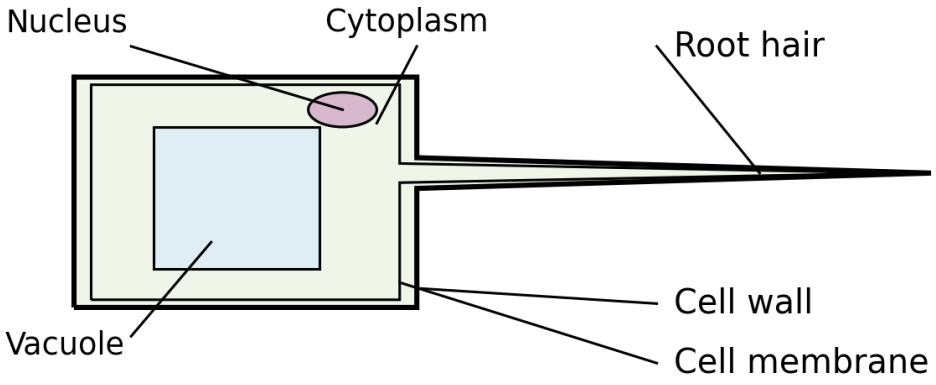
7Ad Cell parts and specialised cells

Book page 17

Cell part	Function	Animal cell	Plant cell
Cell membrane	Controls what enters and leaves the cell.	Yes	Yes
Cytoplasm	Where many chemical reactions happen.	Yes	Yes
Nucleus	Controls cell activities.	Usually	Usually
Mitochondria	Where aerobic respiration releases energy.	Yes	Yes
Cell wall	Supports and strengthens the cell; made of cellulose in plants.	No	Yes
Large permanent vacuole	Contains cell sap and helps keep the cell firm.	No	Usually
Chloroplasts	Contain chlorophyll and carry out photosynthesis.	No	In green cells

Root hair cells

A root hair cell has a long extension that increases its surface area for absorbing water and mineral salts. The root hair is part of the cell.



Root hair cells usually have no chloroplasts because they are underground and do not photosynthesise. Heart muscle cells are specialised to contract and work together to pump blood.

Book answers for 7Aa

Book pages 5 to 7

Page 5 Doctors past and present

1a Symptoms.

1b A cold is a suitable suggestion. The symptoms alone do not prove the illness.

2 Spots or pimples, often with red or inflamed skin.

3a A: an organ.

3b The heart pumps blood around the body.

Page 6 Life processes

1 The completed table is:

Organism	Not an organism
Cow, daffodil, goldfish, mouse, octopus, snake	Car, chair, coal, robot, rock, Sun

2 MRS GREN: Movement, Reproduction, Sensitivity, Growth, Respiration, Excretion and Nutrition.

3 Most animals move from place to place; most plants stay in one place but move parts of themselves.

4 Many plants produce seeds. Animals do not produce seeds.

Page 7 Life processes

5 Any two suitable examples: pulling away from a hot object; turning towards a sound; noticing light; shivering when cold.

6 Trees can continue growing throughout life; humans usually stop increasing in height after reaching adulthood.

7 Most fish absorb dissolved oxygen from water through their gills. Humans take oxygen from air using their lungs.

8a A car can move, use fuel to release energy and give off waste gases. These resemble some activities of organisms.

8b A car does not carry out all seven life processes as a living thing. For example, it cannot grow biologically or reproduce, and it is not made of living cells.

Book answers for 7Ab

Book pages 8 and 9

Page 8 has no numbered questions. The human organ table in the notes summarises its diagram.

Page 9 Human and plant organs

1 Any five correctly matched organs and functions. For example:

Organ	Function
Heart	Pumps blood.
Lungs	Exchange oxygen and carbon dioxide.
Stomach	Churns and digests food.
Small intestine	Digests food and absorbs nutrients.
Kidneys	Remove wastes from blood and make urine.

2 The lungs.

3 The mouth, oesophagus, stomach and small intestine help food enter the body and be digested; the small intestine absorbs nutrients. The large intestine absorbs water. The liver also helps digestion.

4 The lungs remove carbon dioxide, and the kidneys remove wastes in urine. The liver also removes some wastes through bile. Skin excretes small amounts of waste in sweat.

5 The rectum stores faeces; the bladder stores urine.

6 The leaf, because it makes food by photosynthesis.

7a Photosynthesis.

7b Without enough light, the leaves carry out less photosynthesis and make less food. Less food is available for growth and storage in the potato.

8 A suitable answer is the small intestine: both a leaf and the small intestine help provide nutrition. A leaf makes food, whereas the small intestine absorbs digested nutrients. Humans have no organ that makes food by photosynthesis.

For question 8, another comparison can be justified: lungs and leaves both exchange gases. The explanation must match the function being compared.

Remember: removing undigested food as faeces is egestion; removing wastes produced by body activities is excretion.

Book answers for 7Ac Tissues

Book pages 12 and 13

Page 12 Tissues

1 A microscope magnifies objects, making small structures appear larger so details can be examined.

2 The heart is roughly cone-shaped, broad at the top and narrower at the bottom. It has reddish muscle, pale or white fatty areas and blood vessels on the surface.

Page 13 Animal tissues

3a Muscle tissue and fat tissue.

3b Muscle contracts to pump blood. Fat helps cushion and protect the heart.

4 The fatty liver has pale or yellowish, patchy areas consistent with fatty material. It looks less uniformly reddish-brown than the healthy liver shown on page 8.

5 Muscle tissue, because it contracts to push food along the intestines.

Page 13 Plant tissues

6a A storage organ: specifically, a storage root.

6b More than one tissue, at least two. The carrot contains xylem and other tissues, including food-storage tissue. An organ is made of different tissues.

7a Root hair tissue and xylem tissue.

7b Root hair tissue absorbs water and mineral salts from the soil. Xylem carries water and dissolved mineral salts through the plant.

8 The stem. Leaves also contain xylem in their veins.

9 Nutrition: xylem supplies water needed for photosynthesis and carries dissolved mineral salts needed for growth.

Book answers for 7Ac Microscopes

Book pages 14 and 15

Page 14 Using a light microscope

1 Two types of lens: an eyepiece lens and objective lenses.

2a Safety examples: carry it with two hands; keep it away from the edge of the bench; do not let the objective touch the slide; never point a mirror at the Sun; use fine focus at high power.

2b Care examples: use lens paper to clean lenses; keep fingers off lenses; handle slides carefully; cover the microscope after use; return it to low power before storing it.

3 The focusing wheels. The fine focusing wheel sharpens the image, especially at higher magnification.

4 A specimen is the object or sample being examined.

Page 15 Magnification and slides

5 Total magnification = 10 × 15 = ×150.

6 It must be thin so light can pass through it.

7 A coverslip keeps the specimen flat and in place and helps prevent it drying out.

8 Plant examples: onion epidermis and leaf epidermis. Animal examples: cheek lining and skin tissue. Use approved samples or prepared slides as instructed by the teacher.

9 One suitable investigation plan is:

1. Place a drop of water on a clean slide.

2. With teacher supervision, use forceps to obtain a very thin piece of rhubarb stem tissue.

3. Lay it flat in the water; use an appropriate stain if provided.

4. Lower a coverslip slowly at an angle to avoid air bubbles.

5. Start with the lowest-power objective, adjust the light and focus. Increase magnification and use fine focus if needed.

6. Draw and label the structures you can see and record the total magnification. Repeat with another thin piece to check the observations.

Handle glass carefully and wear eye protection as instructed. Any cutting should be done or supervised by the teacher.

10 Possible causes: the lamp is off; the mirror is not directing light upwards; the diaphragm is closed; the slide is not over the stage opening; the objective is not clicked into position; the specimen is too thick; a lens cap or another object blocks the light.

Book answers for 7Ad Cells

Book pages 16 and 17

Page 16 Cells and microscopes

1 A cell is the basic unit from which living organisms and their tissues are made.

2 Granville used a more advanced microscope with higher magnification and clearer detail than Hooke's microscope.

3 Cells. Organisms are made of cells; things that have never been alive are not made of biological cells.

Page 17 Animal cells

4a Nuclei. Each dark blob is a nucleus.

4b They control the activities of the cells.

4c Cell membrane, cytoplasm and mitochondria.

4d The membrane controls what enters and leaves. Many cell reactions take place in the cytoplasm. Mitochondria release energy through aerobic respiration.

Page 17 Measuring cells

5a Measure the widest part of the cheek-cell photograph C, then divide that measurement by 600.

Real width in mm = measured photograph width in mm ÷ 600

Example only: if the photograph measures 54 mm, real width = 54 ÷ 600 = 0.09 mm, or 90 micrometres.

5b Measure the length of a whole plant cell in photograph E, then divide that measurement by 275.

Real length in mm = measured photograph length in mm ÷ 275

Example only: if the photograph measures 55 mm, real length = 55 ÷ 275 = 0.20 mm, or 200 micrometres.

Use the photographs, not the drawings. Your measured values may differ because photocopying, scanning or printing can resize the page. The numbers above illustrate the method rather than fix one answer for every copy.

Book answers for 7Ad Plant cells

Book page 17 continued

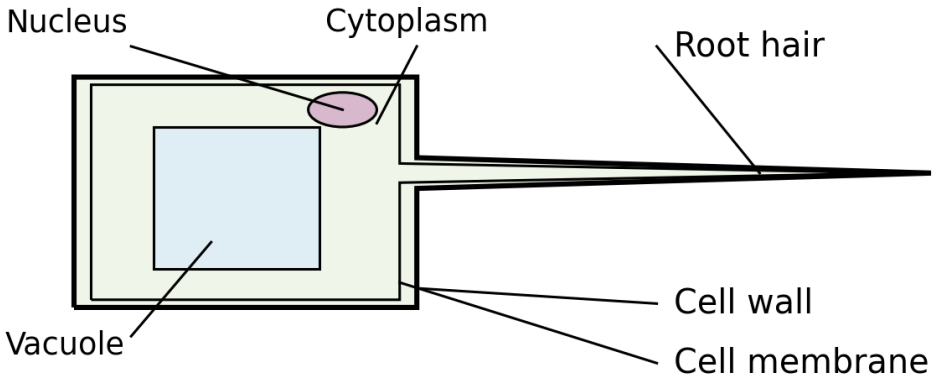
6 A suitable comparison table is:

Part	Typical animal cell	Typical green plant cell
Cell membrane	Present	Present
Cytoplasm	Present	Present
Nucleus	Present	Present
Mitochondria	Present	Present
Cell wall	Absent	Present
Large permanent vacuole	Absent	Present
Chloroplasts	Absent	Present in green cells

7a Chlorophyll inside chloroplasts makes some plant cells green.

7b Chloroplasts are generally bigger than mitochondria. Chloroplasts are visible in the leaf-cell photograph at ×275; the much smaller mitochondria are not visible in these light microscope photographs.

8 A labelled root hair cell is shown below. Label the cell wall, cell membrane, cytoplasm, nucleus, vacuole and root hair extension.



The long root hair increases the surface area for absorption. It normally has no chloroplasts.

Book reference Exploring Science 7 Pages 5 to 9 and 12 to 17